

Evaluation of Deep Learning Models for Tomato Leaf Disease Classification

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Abstract—Tomato leaf diseases can reduce crop productivity and quality when they are not detected at an early stage. Manual disease identification is time-consuming and may be inconsistent because several diseases exhibit similar visual symptoms. This study evaluates six deep learning architectures for tomato leaf disease classification and detection: MobileNetV4, ResNet-18, RepViT, Inception-ResNetV2, YOLOv11, and YOLOv26. The study uses the Roboflow Tomato Leaf Disease Dataset containing 14,522 images distributed across disease and healthy-leaf categories. The images are divided into training, validation, and testing subsets and undergo resizing, normalization, and augmentation before model training. Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix analysis, and, for the YOLO-based models, mean Average Precision (mAP). Inception-ResNetV2 achieves the highest reported overall accuracy of 99.04%. ResNet-18 and RepViT achieve an average precision of 98.39%, while YOLOv26, YOLOv11, and MobileNetV4 achieve average precision values of 97.80%, 96.68%, and 96.63%, respectively. The results show that model selection involves a trade-off between predictive performance, computational efficiency, and output capability. Inception-ResNetV2 is suitable when classification performance is prioritized, MobileNetV4 provides a lightweight alternative for resource-constrained deployment, and YOLO-based models provide additional disease-region localization. The study provides a comparative evaluation of different architectural families using a common tomato leaf disease dataset.

Keywords— Tomato, Leaf Disease, Deep Learning, Image Classification, Object Detection, Smart Agriculture.

Tomato Yellow Leaf Curl Virus, and Powdery Mildew. Early and accurate disease identification is therefore important for supporting timely crop management. However, manual diagnosis requires expertise and can be affected by visual similarity between disease symptoms.

Recent advances in deep learning have improved automated plant disease recognition, particularly through convolutional neural networks and lightweight architectures. Several studies have explored models such as ResNet, MobileNet, EfficientNet, and YOLO for tomato leaf disease classification and detection. However, differences in datasets, preprocessing methods, evaluation of metrics, and experimental settings make direct comparisons between architectures difficult. In addition, high predictive performance does not necessarily indicate efficient deployment, particularly for applications with limited computational resources.

This study presents a comparative evaluation of six pretrained deep learning architectures, namely RepViT, MobileNetV4, Inception-ResNetV2, YOLOv11, YOLOv26, and ResNet-18, for tomato leaf disease classification and detection. The models are evaluated under a consistent experimental workflow covering dataset preprocessing, augmentation, model training, predictive performance, and computational characteristics. The study compares classification and detection performance while also considering model size, parameter count, inference time, and processing throughput to examine the trade-off between predictive accuracy and deployment efficiency. The results provide a practical comparison of different architectures for tomato leaf disease recognition under the evaluated experimental conditions.

I. INTRODUCTION

Tomatoes are an important agricultural crop, but its production can be affected by various leaf diseases that reduce plant health and yield. Common diseases include Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Tomato Mosaic Virus,

II. LITERATUR REVIEW

Deep learning has been widely used for plant disease recognition because convolutional neural networks can extract visual characteristics such as lesions, discoloration, and texture patterns directly from image data [11], [12]. Previous studies have reported strong performance using residual and multi-scale architectures for plant disease identification [13], [14]. Lightweight architectures such as MobileNet reduce computational requirements and are therefore relevant for mobile and edge devices [15].

Other architectures provide different feature representation mechanisms. Vision Transformer models use self-attention to capture relationships between image regions and have been investigated for tomato leaf disease recognition [16]. YOLO-based models address a different requirement by combining disease recognition with spatial localization, making them suitable for applications where the infected region must also be identified [17].

The differences between these architectures indicate that model selection should consider more than predictive accuracy. High-capacity models may provide stronger feature representation but require greater computational resources, while lightweight models prioritize efficiency. Object-detection models provide additional localization information but involve a different evaluation objective from image classification. These considerations motivate a comparative evaluation of the architecture used in this study.

III. METHODOLOGY

3.1 Research Approach

This study applies an experimental research approach to evaluate the performance of several deep learning architectures for tomato leaf disease classification and detection using computer vision techniques. The research mainly focuses on dataset preparation, image preprocessing, model training, and comparative evaluation to identify the most effective model for real-time tomato leaf disease recognition. The dataset used in this research is obtained from <https://app.roboflow.com/aol-ai-2nwdp/tomato-leaf-disease-q4df7-3jjdy/6> [18], which contains annotated tomato leaf images representing multiple disease categories and healthy leaves. The dataset is suitable for image classification and object detection experiments because it provides labeled images with various environmental conditions, lighting variations, and disease patterns. Previous studies have demonstrated that tomato leaf disease datasets are effective for benchmarking deep learning architectures in agricultural image classification and detection tasks.

3.2 Dataset Collection

The dataset used in this study was obtained from the Roboflow Tomato Leaf Disease Dataset. It contains images representing Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Tomato Mosaic Virus, Tomato Yellow Leaf Curl Virus, Powdery Mildew, and Healthy leaves. The images exhibit variations in leaf orientation, illumination, background, and visible disease patterns. The dataset screening process categorizes images with poor quality, irrelevant content, or missing labels into the invalid class. After preprocessing and filtering, 8,617 valid images were used for the experiment.

The valid dataset was divided into training, validation, and testing subsets using a 70:20:10 ratio, resulting in 6,051 training images, 1,696 validation images, and 870 testing images, respectively. The training set was used for model learning, the validation set for monitoring training and model selection, and the testing set for final evaluation. Although the dataset provides variations in visual conditions, it may not fully represent uncontrolled agricultural environments. Therefore, the results should be interpreted as performance on the selected dataset rather than direct evidence of generalization across all field conditions. Similar dataset splitting strategies have been widely applied in tomato leaf disease classification studies [19].

3.3 Data Preprocessing

Prior to the training process, all images undergo preprocessing to improve image quality and enhance model performance. First, all tomato leaf images are resized into 224×224 pixels to match the input dimensions required by the deep learning architectures. Pixel normalization is then applied by scaling pixel values into the range of 0 to 1 to improve numerical stability and accelerate convergence during training. In addition, several data augmentation techniques are implemented to reduce overfitting and increase dataset diversity. These augmentation methods include 15° shear, 90° rotation, flipping, -5% to $+5\%$ exposure. Data augmentation is widely used in plant disease classification research because it improves model robustness under varying environmental conditions and helps models generalize better to unseen data [20].

3.4 Computational and Deployment Evaluation

In addition to predictive performance, the computational characteristics of the evaluated models are assessed for practical deployment using a Lenovo LOQ laptop equipped with an Intel Core i7-13650HX processor, NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB VRAM, and 16 GB RAM running Windows 11 64-bit. The evaluation considers model parameters, model size, inference time, and processing throughput using batch size one for single-image inference, with model warm-up excluded from inference measurements. The computational results are compared with predictive performance to assess the trade-off between model accuracy and computational cost across different deployment scenarios.

3.5 Deep Learning Model Training

This research compares several deep learning architectures for tomato leaf disease classification and detection, including MobileNetV4, ResNet-18, Inception-ResNet-V2, RepViT, YOLOv11, and YOLOv26. Transfer learning is implemented using pretrained weights to improve training efficiency and classification accuracy. Classification-based architectures are used to predict disease categories from tomato leaf images, while YOLO-based architectures additionally perform infected area localization in real time. The training process utilizes the Adam optimizer with a learning rate of 0.001 and applies early stopping to prevent overfitting by monitoring validation loss during training. Previous studies have shown that lightweight architectures and transformer-based approaches can significantly improve tomato leaf disease classification performance while

maintaining computational efficiency suitable for mobile deployment [21], [22].

3.5 Performance Evaluation

The trained models are evaluated using several performance metrics to analyze both classification effectiveness and computational efficiency. The evaluation metrics used in this study include accuracy, precision, recall, F1-score, mean Average Precision (mAP), inference speed, and model size. Accuracy measures the overall prediction

correctness, while precision and recall evaluate the model's capability to correctly classify disease categories. F1-score is used to balance precision and recall performance. For YOLO-based architectures, mAP is additionally utilized to evaluate object detection and localization performance. Furthermore, a confusion matrix is generated to analyze classification errors among visually similar disease classes such as Early Blight and Late Blight. Similar evaluation strategies have been widely adopted in recent tomato leaf disease detection studies [23].

3.6 Flowchart

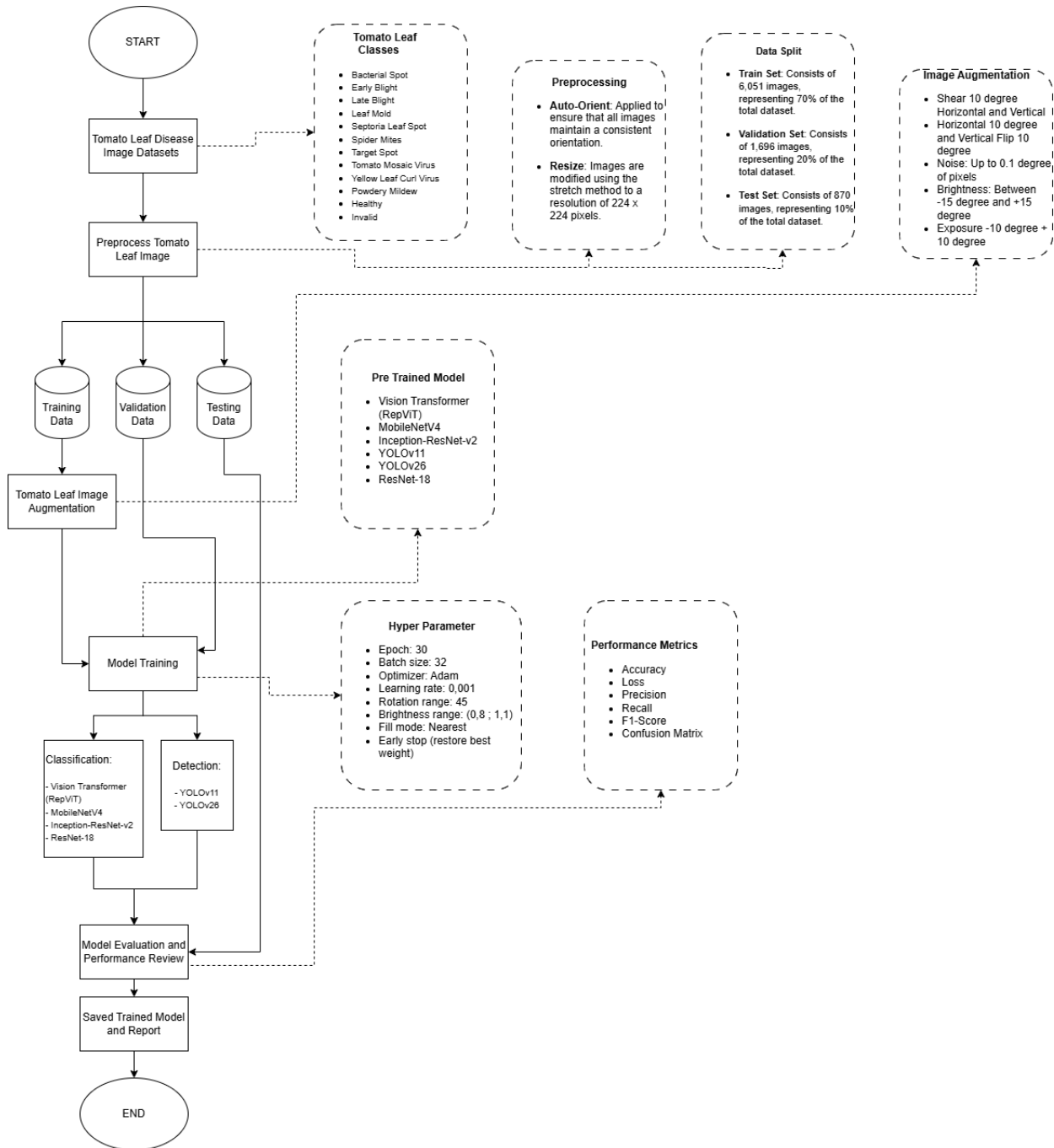


Fig 1. Flow Diagram of Deep Learning Models of Tomato Leaf Disease Classification

The process begins with the collection of the Tomato Leaf Disease Image Dataset, which contains multiple tomato leaf classes, including Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Tomato Mosaic Virus, Tomato Yellow Leaf Curl Virus,

Powdery Mildew, Healthy, and Invalid classes. The images are first preprocessed using automatic orientation to maintain consistent image orientation and resized to 224×224 pixels. The dataset is then divided into training, validation, and

testing subsets, consisting of 6,051, 1,696, and 870 images, respectively, corresponding to a 70:20:10 split.

The training data subsequently undergo image augmentation to increase data variation and improve model robustness. The augmentation configuration includes shear, horizontal and vertical flipping, rotation, noise, brightness adjustment, and exposure adjustment. The augmented training data are then used to train six pretrained deep learning models: RepViT, MobileNetV4, Inception-ResNetV2, YOLOv11, YOLOv26, and ResNet-18.

The trained models are divided into two approaches. MobileNetV4, ResNet-18, RepViT, and Inception-ResNetV2 are used for image classification, while YOLOv11 and YOLOv26 are used for object detection. The training process is configured with 30 epochs, a batch size of 32, the Adam optimizer, and a learning rate of 0.001. Additional training configurations include the specified rotation range, brightness range, nearest fill mode, and early stopping with the best model weights restored.

After training, the models are evaluated using accuracy, loss, precision, recall, F1-score, and confusion matrix analysis. The resulting models are then evaluated and compared based on their performance, after which the trained model and evaluation report are saved as the final outputs of the experimental workflow.

IV. RESULT AND DISCUSSIONS

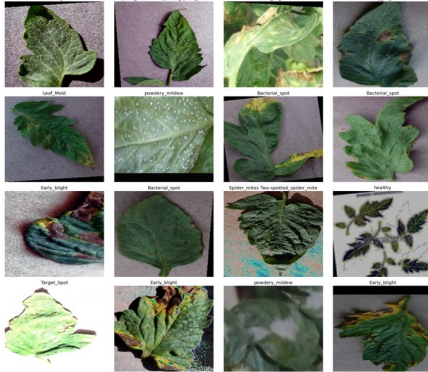


Fig 2. Dataset Tomato Leaves

1. Powdery Mildew

In Fig. 1, the Powdery Mildew class shows strong classification performance. The disease is characterized by a white powder-like layer on the leaf surface, which makes it visually distinct from other classes. Due to these unique features, most models are able to classify this disease accurately.

2. Bacterial Spot

The Bacterial Spot class demonstrates relatively good performance, as the model is able to detect yellowish and dark lesions on the leaf surface. However, in some cases, the similarity between Bacterial Spot and early-stage Early Blight may lead to confusion due to overlapping color patterns.

3. Early Blight

The Early Blight class presents moderate difficulty for the model. As shown in Fig. 1, this disease is characterized by dark spots with concentric rings. However, these features can

sometimes resemble other diseases, particularly Late Blight, making accurate classification more challenging.

4. Late Blight

In Fig. 1, Late Blight exhibits irregular dark regions and leaf decay patterns. The similarity in texture and color distribution with Early Blight contributes to potential misclassification. This indicates that the model may struggle with fine-grained feature differentiation.

5. Leaf Mold

The Leaf Mold class shows relatively stable performance. Its dark green patches and distinct texture help the model differentiate it from other diseases. However, variations in lighting conditions may slightly affect prediction accuracy.

6. Spider Mites

The Spider Mites class is identified by small speckled patterns and leaf discoloration. While the model is able to detect these features, the small size and scattered nature of the patterns may reduce classification consistency.

7. Target Spot

The Target Spot class demonstrates strong classification performance due to its circular lesion patterns, which are visually distinctive. These unique features allow the model to classify this disease with higher confidence.

8. Healthy

Healthy leaves are generally classified correctly due to the absence of disease patterns. However, variations in lighting and background may occasionally affect classification results.

A. Inception-ResNetV2

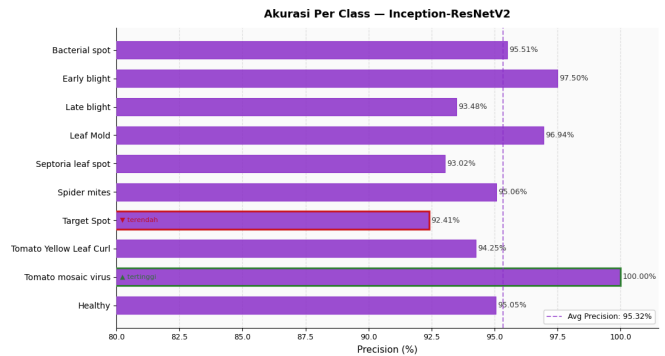


Fig 3. Accuracy per class of Inception-ResnetV2 Model

Inception-ResNetV2 achieves the highest reported overall accuracy of 99.04%. Its combination of multi-scale feature extraction and residual connections is suitable for disease patterns that vary in size and appearance. Small lesions require fine-grained representation, while larger areas of discoloration require broader spatial information. The architecture can capture both types of features through its multi-scale design. However, the higher representation capacity also needs to be considered together with computational requirements when deployment resources are limited.

B. YOLOv26



Fig 4 Accuracy per class of Yolo26 Model

YOLOv11

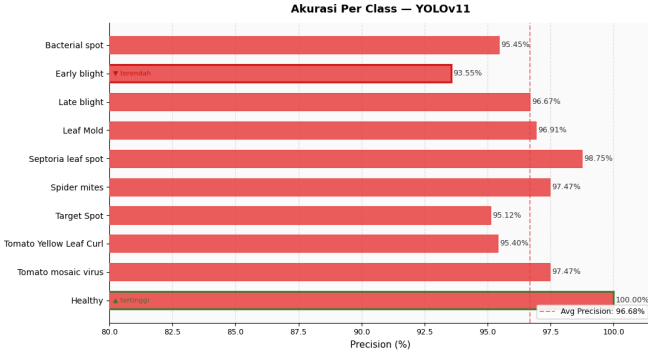


Fig 5. Accuracy per class of Inception YOLOv11 Model

YOLOv11 achieves an average precision of 96.68%, while YOLOv26 achieves 97.80%. Among the evaluated YOLO models, YOLOv26 therefore provides the higher reported precision. In addition to disease recognition, both architectures provide spatial localization of detected regions. This capability is useful for applications where identifying the infected area is required rather than only predicting the disease label.

C. ResNet- 18

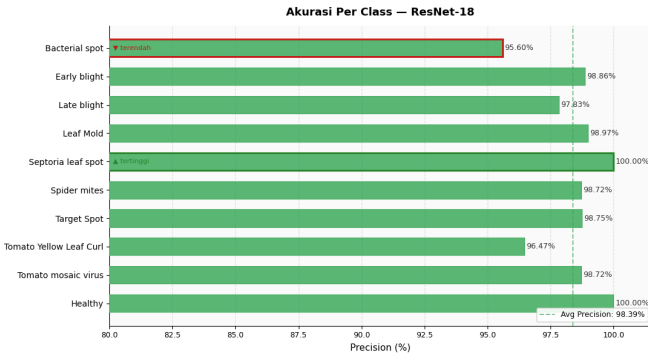


Fig 6. Accuracy per class of Resnet-18 Model

ResNet-18 achieves an average precision of 98.39%. The residual connections help maintain stable feature propagation and support the extraction of disease-related texture and lesion patterns. The result indicates that a relatively compact residual network can achieve strong classification performance without the depth of Inception-ResNetV2.

D. RepVit

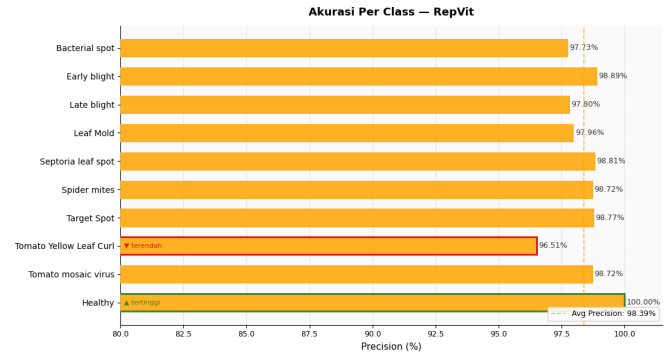


Fig 7. Accuracy per class of RepViT Model

RepViT achieves an average precision of 98.39%. The model maintains stable precision across the evaluated classes, with the Healthy class reaching 100.00% and Tomato Yellow Leaf Curl Virus recording the lowest precision at 96.51%. Its performance suggests that efficient visual feature representation can provide competitive results for tomato leaf disease classification.

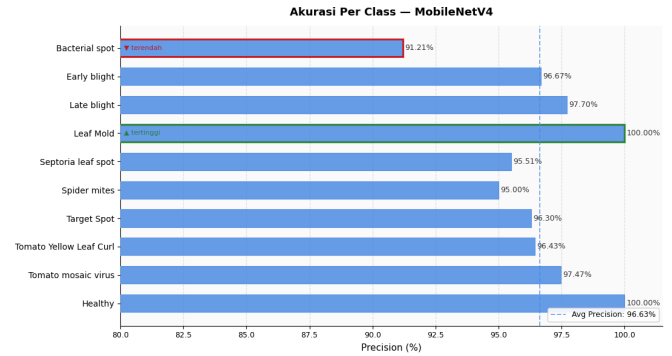


Fig 8. Accuracy per class of MobileNetV4 Model

MobileNetV4 achieves an average precision of 96.63%. Leaf Mold and Healthy reach 100.00% precision, while Bacterial Spot records the lowest precision at 91.21%. The lower performance on Bacterial Spot is consistent with the visual similarity between some lesion patterns and other disease classes. The lightweight design makes MobileNetV4 a suitable candidate for resource-constrained deployment, although it does not achieve the highest reported predictive performance.

V. CONCLUSION

This study evaluated six deep learning architectures for tomato leaf disease classification and detection using the Roboflow Tomato Leaf Disease Dataset. Inception-ResNetV2 achieved the highest reported overall accuracy of 99.04%, while ResNet-18 and RepViT achieved an average precision of 98.39%. MobileNetV4 achieved 96.63% average precision and provides a lightweight alternative for resource-constrained environments. YOLOv11 and YOLOv26 achieved 96.68% and 97.80% average precision, respectively, while also providing disease-region localization.

The comparison shows that the architectures provide different advantages. Inception-ResNetV2 is suitable when classification performance is prioritized, MobileNetV4 is more appropriate when computational resources are limited, and YOLO-based models are useful when disease localization

is required. The results also show that visually similar diseases, particularly Early Blight and Late Blight, remain challenging to distinguish.

The main contribution of this study is a comparative evaluation of classification-oriented and object-detection-oriented architectures under a common tomato leaf disease dataset. However, the results are limited by the characteristics of the selected dataset and may not fully represent uncontrolled field conditions. Future work should evaluate the models using external field datasets and further test their performance on mobile and edge hardware.

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AUTHOR CONTRIBUTION

Abdul Harris Rangkuti supervised the research design and model selection. Fa'iq Haidar Prasetya conducted the experiments and dataset preparation. Nicholas Axel Nathaniel Sugiarito and Stefany Go contributed to the data collection, implementation, analysis, and manuscript preparation. All authors reviewed and approved the final manuscript.

DATA AVAILABILITY

An overview of the availability of Tomato Leaf Disease Classification datasets for the image classification can be found at link <https://app.roboflow.com/aol-ai-2nwdp/tomato-leaf-disease-q4df7-3jjdy/6>